

Housing estates and the 2025 vote in Prague's precincts

Education and citizenship explain 43 % of the rise in ANO's 2025 vote with the share of panel flats, short of the registered half (not supported); the gradient steepened after 2017 (supported).

ANO, PRECINCTS WHERE MORE
THAN SEVEN IN TEN FLATS ARE IN
PANEL BUILDINGS

16 % where fewer than one in ten are · administrative count

26 %

ANO'S ESTATE GRADIENT
EXPLAINED BY EDUCATION
AND CITIZENSHIP
(REGISTERED BLOCK)

range over 48 specifications 31–44 % · 282 clusters in 56 districts

43 %

ANO'S
ESTATE
GRADIENT,
2017 →
2025

pp per 10 pp of panel flats · change +0.26, range over matching rules +0.24 to +0.30 · 819 precincts in 49 districts

0.55 → 0.80

THE
METRO,
PER
DOUBLING
OF
DISTANCE

90 % CI -0.40 to +0.05 · not supported: inconclusive · 282 clusters in 56 districts

–0.17 pp

OVERWHELMED? HERE'S THE SHORT VERSION

- ANO does better the more of a precinct's flats are in panel buildings and the further it lies from the centre, while the Pirates fall with the distance from the centre and SPOLU with the share of panel flats. Once housing and the centre are known, the distance to the nearest metro station adds nothing that can be told from zero, and the registered test could not show that its effect is negligible either.
- The registered composition block leaves out age, because the estates' age structure partly follows from how their flats were allocated. With age added, composition explains about half of ANO's estate gradient, and the test can then no longer tell the answer from one half.

- The steepening rests on precincts whose boundaries stayed the same, which hold more panel flats than the city as a whole, and it holds housing fixed but not residents, so it may partly reflect who moved into the estates or aged there.
- The two turnout tests are uninformative: the register count the Statistical Office publishes includes voters who came with a voting pass, so these files cannot show whether registration explains the centre's low turnout.
- Every measurement is of areas, so nothing here says how any resident votes, and none of the estimates is a causal effect of living in a panel building.

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Housing stock or residents

In the Chamber election of 3–4 October 2025 Prague voted SPOLU 33.9 %, ANO 19.9 %, the Pirates 16.9 % and STAN 13.4 %, on a turnout of 71.4 %¹, and across the city's 1 119 precincts ANO did best in the panel housing estates. Part 1 of this series followed the city council's votes; this part follows the voters. A panel estate is both a housing stock and a population: people who were allocated flats there in the 1970s and 1980s and grew old in them, their children, and later buyers of the cheapest flats in the city. ANO may do better in the estates because of the estates themselves (their distance, their shops, their neighbours) or because the estates hold more of the people who vote ANO anywhere. Electoral geography has argued about this for four decades, as context against composition^{2,3}, and for immigration attitudes in large European cities the best recent panel evidence finds that most of what looks like place is people⁴. Aggregate data cannot settle the question for individuals^{5,6}, but they can measure how much of the estates' gradient is left once what the census knows about their residents is held equal. Prague adds a post-socialist case to that literature: its estates were built by the state and their flats allocated administratively rather than through a housing market, and the cadastre records the construction of every residential building, so housing can be measured building by building. A follow-up study, its design written and reviewed before modelling, therefore adds the 2021 census and the elections of 2017 and 2021 to the 2025 precincts and asks four things: how much of ANO's estate gradient is explained by who lives in the estates, whether the metro's effect is small enough to be called negligible, whether the estates' vote has moved over eight years, and why turnout is lowest in the centre.

Data and the 2025 precinct map

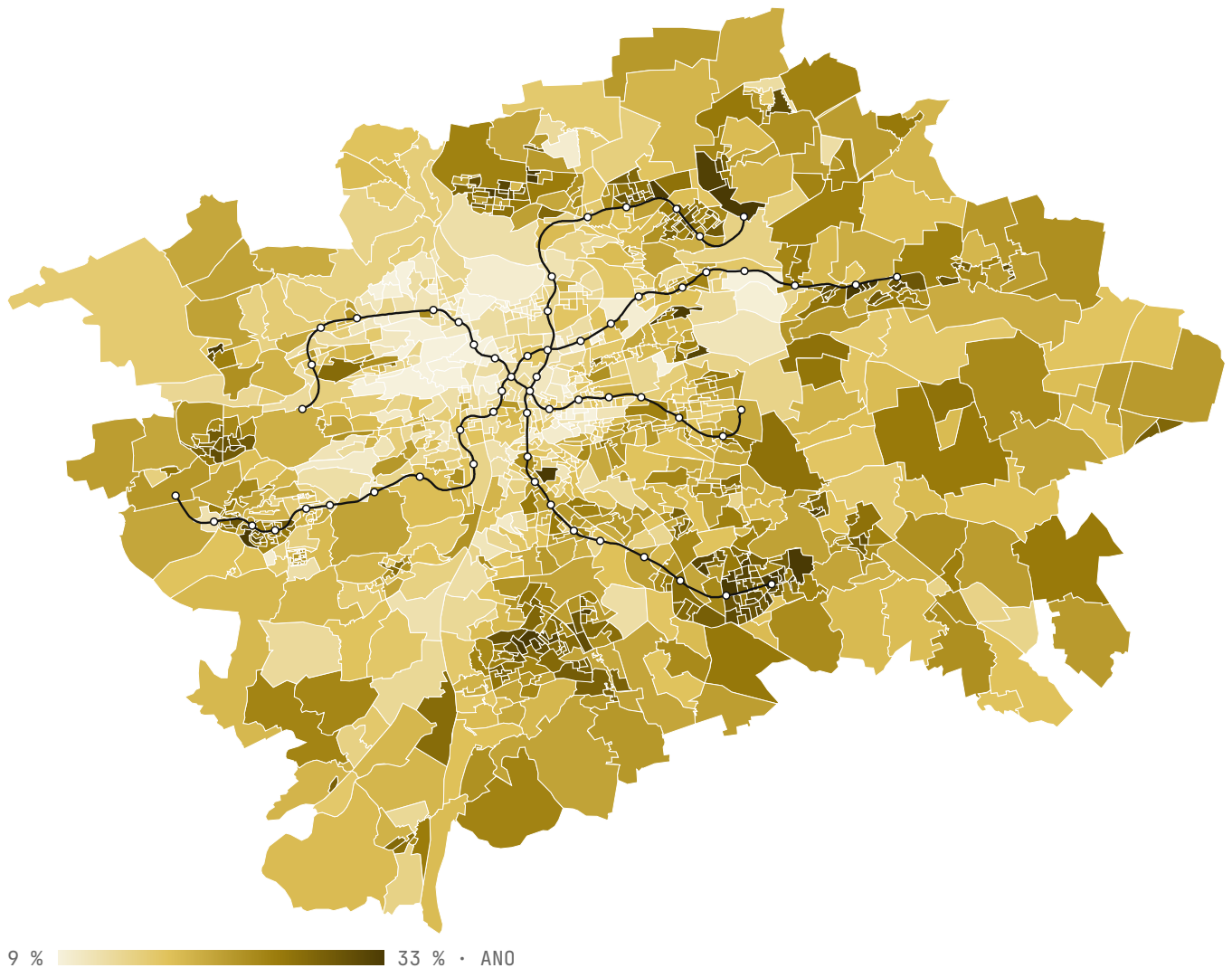


fig. 1 · Share of valid votes by precinct, Chamber election 2025, 1 119 precincts (ANO unless you pick another party, turnout or the share of panel flats above the map; hover or tap a precinct for its numbers); black: the metro and its 58 stations as they ran on election day. The darkest patches are housing estates: of the 40 precincts where ANO did best, 13 are in Praha 11 (Jižní Město) and most of the rest in Praha 8, 14, 12 and 18. Sources: ČSÚ, ČÚZK RÚIAN, PID GTFS, accessed 28 September 2026.

ANO's share rises with distance from the centre, from 13 % within two kilometres of Můstek to 25 % between eight and ten, then falls back to 22 % beyond ten, where the estates give way to family houses and villages inside the city boundary. The Pirates run the other way, from 22 % in the centre to 13 % at the edge, while SPOLU, the largest party, moves least, from 37 % to 31 % and back to 34 %. Three measurements can be put on every precinct: how far its centre is from the nearest metro station, how far it is from Můstek, and what share of its flats are in panel buildings. Of the 735 755 flats in the precincts, 32 % are in panel buildings. Figure 2 lays the three side by side on one scale.

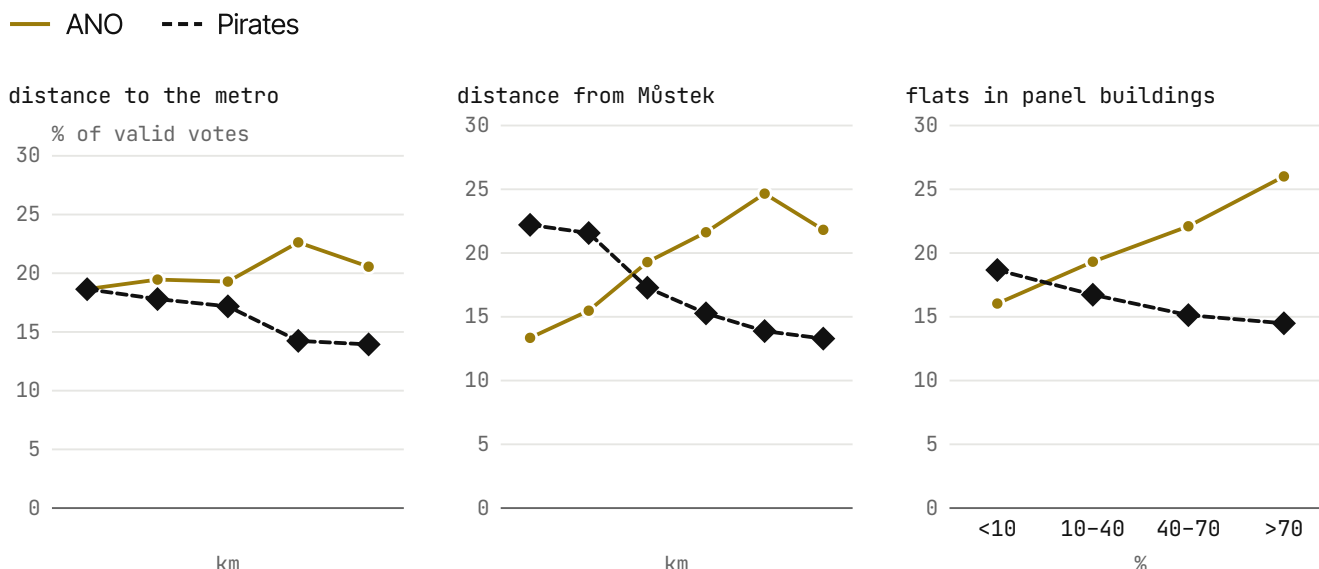


fig. 2 · ANO (ochre) and the Pirates (black, dashed), % of valid votes, grouped by each measurement. Distances are from the precinct's centre; Můstek stands for the centre; a panel building is one the cadastre records as built of wall panels. Two precincts with no flats are left out (1 117 of 1 119 remain). Sources: ČSÚ 2025 precinct results; ČÚZK RÚIAN; PID GTFS; accessed 28 September 2026.

Near the metro or far from it, ANO's share stays between 19 and 23 %, and the small rise that exists comes with distance from the centre, because the stations thin out as the city does. Across the share of panel flats the change is much larger: where fewer than one flat in ten is in a panel building, ANO takes 16 % and SPOLU 37 %; where more than seven in ten are, ANO takes 26 % and SPOLU 29 %. A third of the city's flats are in panel buildings, so the gradient concerns a large part of the city rather than its margins.

In one regression weighted by valid votes, with the three measurements and the share of flats in family houses as a fourth, the metro's coefficient for ANO is -0.05 pp per doubling of the distance to the nearest station (95 % CI -0.28 to $+0.19$), which cannot be told from zero. Every ten percentage points of panel flats go with 0.9 pp more for ANO (0.8 to 1.0), and every doubling of the distance from Můstek with 2.0 more (1.5 to 2.4). The Pirates' share falls by 2.0 pp for every doubling of the distance from the centre and shows no measurable metro effect either. The model explains 56 % of the variation in ANO's share across precincts, and 54 % of the Pirates'. SPOLU loses 0.6 pp for every ten points of panel flats and gains 0.5 (0.4 to 0.7) for every ten points of flats in family houses, which is why it recovers beyond ten kilometres, in the villa quarters and villages of its outer ring. The smaller parties follow the three measurements only weakly, and for them the model explains 13–29 % of the variation. These 2025 estimates were known before the study below was designed.

The 2025 precinct counts add up to the Statistical Office's published result for Prague: 1 119 precincts, 898 829 registered voters, 632 938 valid votes, SPOLU 33.94 %, SPD 5.24 %¹. Boundaries come from the cadastre's current register, which keeps only the latest version of each precinct². Every record in it is dated after the election, most from two bulk updates in October and November 2025, and 22 were changed from December on; dropping those 22 moves no estimate by more than a tenth of a point. The metro stations come from the public-transport timetable⁸, which lacks Flora, closed for reconstruction since 2 February 2026⁹, so the station is added back as it ran on election day. A building's construction type is recorded for 98 % of the flats, and "panel" here means the cadastre's wall panels, concrete included.

The census counts people on a grid of 1 km squares¹⁰. Prague's median precinct is an eighth of a square, so every cell's counts were spread over the precincts it overlaps, in proportion to the flats standing in each on

census day, from the cadastre's building register. This dasymetric allocation¹¹ puts 1 301 482 residents into Prague's precincts against the census's 1 301 432, and an independent allocation from the census's 958 basic settlement units in Prague agrees within 25 % for all but 13 precincts.

How this was done

The design fixes the questions, the hypotheses, the one test for each, the decision rules and fourteen robustness checks¹². It was written and reviewed before modelling, but it first appears in the repository in the same commit as the results, so its timing is not independently timestamped. When it was written, the 2025 estimates above were known, and so were the column headers and marginal totals of the census grid; no census count or earlier election had been joined to a precinct. Because the 2025 estimates were known, the metro hypothesis is framed as a test of whether the absence of a measurable effect in 2025 holds up as a negligible effect. Everything that changed afterwards is listed below.

A precinct gets only the composition of the square it sits in, while its housing is measured building by building. Comparing the two at different resolutions would flatter housing, so the main models add precincts up into the square that holds most of their flats, which gives 282 clusters, each measured at one resolution. The regressions weight by valid votes. Uncertainty comes from a wild cluster bootstrap by city district^{13,14}, with conventional, small-sample-corrected¹⁵ and spatial¹⁶ standard errors computed alongside. Each cluster belongs to the district that holds most of its flats, and the 282 clusters fall in 56 of Prague's 57 districts, so the effective sample for the composition and metro tests is 56 groups rather than 282 observations; the steepening test uses 819 precincts in 49 districts and the turnout tests 250 clusters in 44. The six confirmatory tests share a 5 % error rate through Holm's procedure¹⁷.

The registered hypotheses and their results

Seven hypotheses were registered. H1a checks that the census measure of education behaves as expected and sits outside the confirmatory family; of the six inside it, one is supported after Holm's correction, and the two turnout tests are uninformative.

Hypothesis	Test	Result
H1a · ANO does worse where more adults have a university or higher vocational degree check that the census measure works	coefficient < 0	supported -0.33 pp per pp, p < 0.001
H1b · education and citizenship explain more than half of ANO's estate gradient	$\theta > 0.5$	not supported $\theta = 0.43$, p = 0.97
H1b' · the same for SPOLU's (negative) estate gradient	$\theta > 0.5$	not supported $\theta = 0.42$, p = 0.91
H2 · the metro's effect on ANO is within ± 0.25 pp per doubling of distance	equivalence (TOST)	not supported: inconclusive p = 0.28
H3 · ANO's estate gradient was steeper in 2025 than in 2017	interaction > 0	supported +0.26 pp per 10 pp, p < 0.001
H4a · the centre's turnout gap is smaller counted against residents than against the register	gradient smaller	uninformative p = 0.80, sign reversed
H4b · registrants who are not residents add less than one voter each	elasticity < 1	uninformative 0.99 ± 0.02 , p = 0.37

θ : share of the estate gradient accounted for by the composition block. p-values: one-sided, wild cluster bootstrap by district, 9 999 draws (56 districts for H1 and H2, 49 for H3, 44 for H4); H2 uses the largest of three

standard errors. Holm thresholds run from 0.008 to 0.05; only H3 passes. H1a is outside the family. H4a and H4b were registered as tests of the register and were relabelled uninformative after estimation, because the published register count includes voting-pass voters; their p-values are shown as computed.

How much of the estate gradient is composition

Across the 282 clusters, ANO's share rises by 0.99 pp for every ten percentage points more flats in panel buildings, holding the distance from the centre, the distance to the metro and the share of family houses equal. Adding what the census knows about education and citizenship lowers the gradient to 0.56 pp (95 % CI 0.47 to 0.66, bootstrap by district), and the census block raises the share of variation explained from 70 % to 87 %.

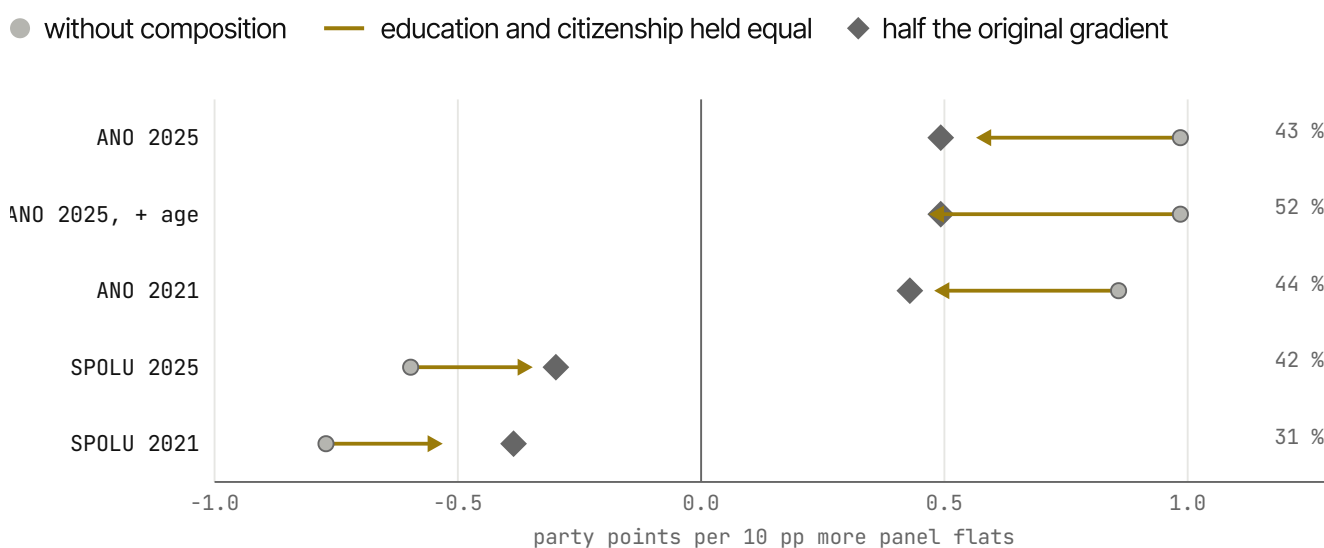


fig. 3 · The estate gradient, in party pp per 10 pp more panel flats, before (light grey) and after (ochre arrow head) holding education and citizenship equal; the grey diamond is half the original gradient, the line H1b had to cross. Right: the share explained. 282 grid clusters (280 in 2021), weighted by valid votes. Sources: ČSÚ elections 2021 and 2025, Census 2021 grid, accessed 28 September 2026.

A decomposition of the gap¹⁸ puts 0.44 of the 0.42 points on education and nothing on the share of foreign citizens, so education accounts for almost all of the fall. Of the gradient, 57 % survives, and the test in the other direction, whether more than half is left, rejects at $p = 0.03$. For SPOLU the pattern is the same and weaker, with 42 % explained and neither direction significant.

The census measures composition over a square kilometre, which is coarser than a street, and at registration this was expected to under-control for composition. The direction is not guaranteed here, because the census counts were spread over precincts in proportion to their flats, so the allocation error is correlated with the housing measure itself; 43 % is therefore likely understated, not a proven lower bound. The registered block also leaves out age, because the estates' age structure is itself partly a product of the housing: flats were allocated to young families in one decade, and those families aged in place. With the share of over-65s and under-15s added, composition explains 52 %, about half, and the test can no longer tell the answer from one half. The 2021 election, a replication specified in the design, gives 44 % for ANO and 31 % for SPOLU.

Distance to the metro

The 2025 precinct model found no measurable effect of the metro on the vote, and an interval around zero does not show that an effect is absent, so the registered test asked the stronger question of whether the metro's effect, in the full model with composition, is small enough not to matter. The threshold was set at a quarter of a point per doubling of the distance to a station. Across the middle half of Prague's precincts that is half a point, a fortieth of ANO's vote in the city.

● estimate and 90 % interval ± 0.25: too small to matter

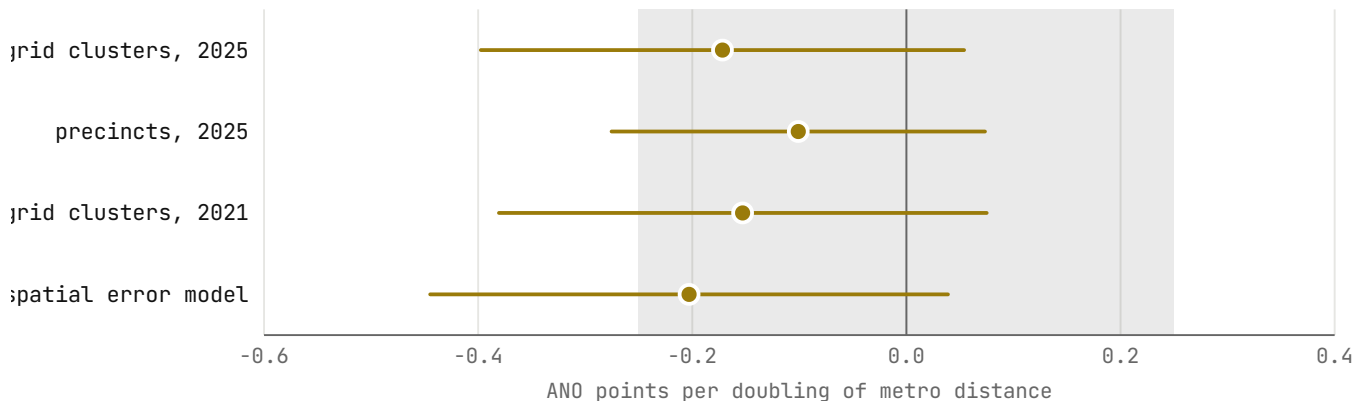


fig. 4 · The metro's coefficient for ANO in pp per doubling of the distance to a station, with its 90 % CI (the interval an equivalence test at 5 % uses), against the grey band declared too small to matter. 282 clusters (2025), 1 117 precincts, 280 clusters (2021). Sources: ČSÚ, Census 2021 grid, PID GTFS snapshot of 28 September 2026.

Every estimate is small and every interval crosses zero, but every interval also reaches past -0.25 . The design-stage power check had predicted this, since with 282 clusters the standard error was bound to be larger than the margin. The margin was not widened afterwards, because the 2025 estimate was already known and a margin chosen after seeing it would prove nothing. The test is therefore inconclusive, because with 282 clusters a negligible metro effect cannot be told from a small one. If the estimates lean anywhere, it is towards ANO doing slightly better closer to stations once the rest is held equal.

ANO's estate gradient in 2017, 2021 and 2025

Precinct boundaries move between elections and the cadastre keeps only the current ones. The comparison over time therefore uses the 819 precincts that exist under the same number in 2017, 2021 and 2025 and whose register grew or shrank no more than 10 % beyond what new flats inside them predict. Matched precincts have more panel flats than the rest (40 % against 27 %), because the unmatched are mostly where the city is still building, so the comparison describes the older city rather than all of it.

— 819 matched precincts ■ 5 % (410) and 20 % (982) matching rules

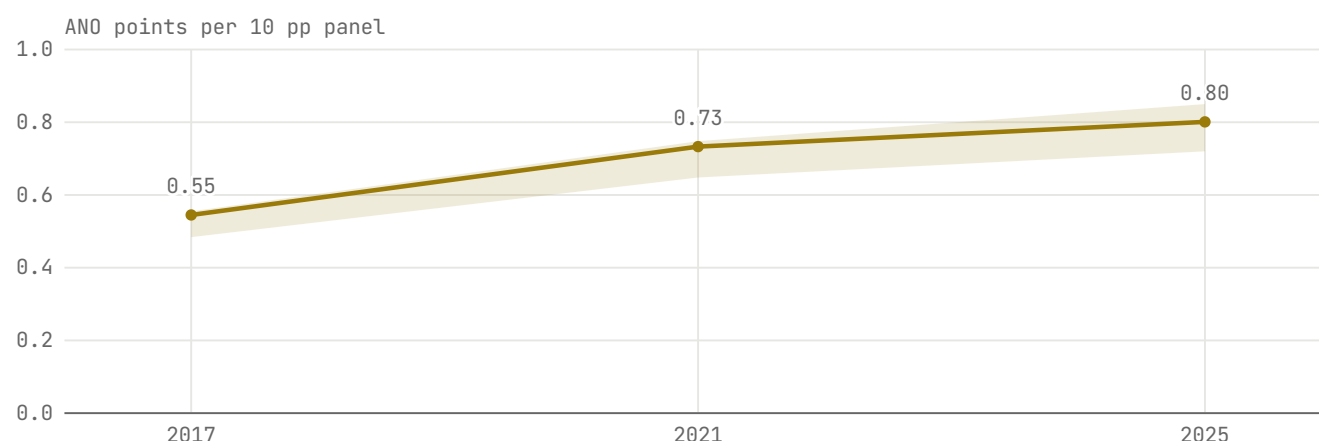


fig. 5 · ANO's estate gradient in pp per 10 pp of panel flats, by election, 819 matched precincts (ochre line); light ochre band: the same with the 5 % (410 precincts) and 20 % (982) matching rules. Distance from the centre, distance to the metro and family-house share are held equal in each year. Sources: ČSÚ elections 2017, 2021, 2025; ČÚZK RÚIAN; accessed 28 September 2026.

Between 2017 and 2025 ANO's estate gradient grew by 0.26 pp per 10 pp of panel flats ($p < 0.001$), from 0.55 to 0.80, with 2021 in between. It is the one registered hypothesis that passes the family-wise threshold. It holds under the stricter and the looser matching rule (+0.24 and +0.30) and with weights that correct for which precincts could be matched (+0.26). The model holds housing fixed but not people, since the census was taken once, between the elections, so the result shows that ANO's vote became more concentrated in the estates without showing whether the estates or their residents changed.

Turnout in the centre

The published register count per precinct includes voters who came with a voting pass, and passes are used most where people are away from home: in all 14 819 precincts in the file, abroad included, envelopes never exceed that count, which is what one would see if pass voters are added to it. This was found only after the data were joined, and because the published files do not allow the register and the pass voters to be separated, both turnout tests are uninformative. In 2025 turnout was lowest in the centre, 66.8 % within two kilometres of Můstek against about 73 % from four kilometres out, and it barely changed with the share of panel flats (71–72 % across the four groups). The registered explanation was that the centre has more voters who are registered there but live elsewhere, since Czech permanent residence is an address in a register and people with no address of their own are registered at the district office. With the 158 precincts that contain a district office, a hospital, a care home or a prison left out, turnout counted against the register rises by 1.9 pp per doubling of the distance from Můstek, while counted against the census's resident Czech adults it falls by 4.3, and envelopes scale with the register at an elasticity of 0.99, with residents adding nothing once the register is known. Both results go against the registration explanation, but with pass voters inside the register count neither can be read as a test of it.

Robustness checks

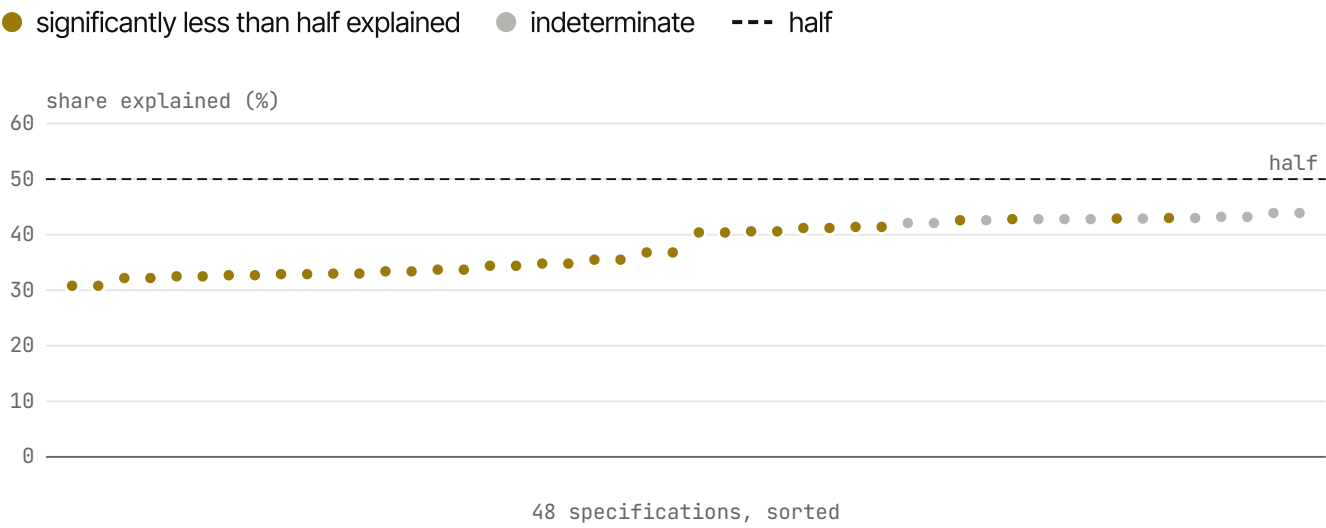


fig. 6 · Share of ANO's estate gradient explained by education and citizenship (the registered block, without age) in 48 specifications: weighted or not, allocation by flats or by area, panel defined narrowly or broadly, all clusters or large precincts only or homogeneous squares only, bootstrap or spatial errors. Ochre: significantly less than half explained. Light grey: indeterminate. None: more than half. Sources as fig. 3.

Check	ANO θ	SPOLU θ	metro
main	43 %	42 %	-0.17
unweighted	34 %	36 %	-0.20
without precincts redrawn after Dec 2025	43 %	44 %	-0.16
precincts with 300+ voters	43 %	42 %	-0.17
panel: wall panels only	43 %	44 %	-0.18
census spread by area	40 %	36 %	-0.14
education stated, 50+ residents	42 %	40 %	-0.18
composition of citizens only	42 %	42 %	-0.19
allocation check passed	41 %	39 %	-0.15
homogeneous squares only	44 %	38 %	-0.43

Further check	Result
48 specifications of the registered block (fig. 6)	ANO θ 31–44 %
leave one district out	θ moves by at most 0.02
fractional logit, for shares between 0 and 1 ¹⁹	ANO θ 46 %
single precincts (composition coarser than housing)	ANO θ 30 %
spatial error model; residual Moran's I = 0.06	same answers, Hausman test ²⁰ p = 0.62
panel shares shuffled within districts	attenuation 0.19 pp, observed 0.42

Across the 48 specifications composition explains between 31 and 44 % of ANO's estate gradient under the registered block. Single precincts give a lower share because their composition is coarser than their housing, which inflates the leftover gradient by construction. Shuffling panel shares within districts leaves an apparent attenuation of 0.19 pp rather than zero, because districts differ in both housing and people, and the observed 0.42 lies outside that range.

What changed after registration

- 28 September 2026, design stage: the rule to publish only after the 2026 municipal elections was dropped. It concerned publication, not analysis.
- 28 September 2026, design stage: the power check found the metro test underpowered from the start. This was recorded then, and the margin was kept.
- 28 September 2026, operational details the design left open: only five-year age groups are published, so 15–17 year olds are estimated as three fifths of the 15–19 group; dormitories are not flagged, because no open list of them exists; the 2017 precinct geometry was not recovered from the cadastre's change files, so the registered matching rule is used alone.
- 28 September 2026, estimation: the wild cluster bootstrap is unrestricted rather than restricted as registered, because the composition contrast spans two regressions and has no standard restricted form. A coding error in H4a, which tested a signed rather than an absolute difference, was found and fixed before reporting.
- 28 September 2026, estimation: the placebo with the planned line D stations was not run, because official station coordinates could not be found in open data.
- 28 September 2026, estimation: the published register count was found to include voting-pass voters. It was first reported as a limitation of the turnout tests; on 29 September 2026 H4a and H4b were relabelled from "not supported" to "uninformative" for that reason.
- 29 September 2026: the claim that 43 % is a lower bound on what composition explains was withdrawn, because the allocation error is correlated with the housing measure; it now reads "likely understated". The bootstrap group count was corrected from 57 to 56 districts, the number the estimation actually used.

What this means

The results locate ANO's vote in the housing estates and measure how much of it goes with who lives there, without identifying any effect of the estates on anyone's vote. Measured are the votes, the housing and the distances: ANO takes 26 % where more than seven flats in ten are in panel buildings and 16 % where fewer than one in ten are. Inferred, from 1 km census squares, is the share of that gradient going with education and citizenship, 43 % under the registered block and 31–44 % across 48 specifications, likely understated, and about half once age is added. The rest cannot be credited to the estates as places, since income, tenure and length of residence are unmeasured. Between 2017 and 2025 the gradient rose by 0.26, from 0.55 to 0.80 pp per 10 pp of panel flats (+0.24 to +0.30 across matching rules), in the older, matched precincts; with residents not held fixed, it shows the vote concentrating in the estates without saying why. The metro's association, -0.17 pp per doubling of distance (90 % CI -0.40 to $+0.05$), is neither distinguishable from zero nor shown to be negligible, and the turnout tests are uninformative.

What this does not show

No result here says how any resident votes, because every measurement is of areas⁵. The census is from March 2021, four and a half years before the 2025 election, taken during distance teaching and before the refugees of 2022. The grid is coarse, which limits what composition can explain. The areas are Prague's, and a different grid

would give somewhat different numbers²¹. The leftover estate gradient is what the census does not measure, and the place itself is only one candidate for it: income, tenure and how long people have lived there are not in the grid, and any of them could carry it. The steepening since 2017 describes the matched, older precincts, with residents' composition not held fixed over time. None of the estimates is a causal effect of living in a panel building.

Reproduction

The analysis runs from `tools/praha/` in the site's repository. Each script's header gives its `uv run` command with its dependencies (pandas, numpy, scipy, statsmodels, shapely, pyproj, pyarrow, matplotlib); no lock file pins their versions yet. Raw downloads are cached in `tools/data/praha2/` and `tools/data/praha2x/`, which are not committed, and the hashes of the files present at registration are in [prague-rings-files.sha256](#). The scripts run in this order:

- [metro.py](#): the 2025 description and the map data
- [rings_data.py](#): the design frame, before any result is joined
- [rings_power.py](#): the design-stage checks
- [rings_extended.py](#): the registered tests, written to `assets/praha/rings_extended.json`
- [rings_robust.py](#): the robustness checks, written to `assets/praha/rings_robust.json`
- [figures_metro.py](#) and [figures_rings_extended.py](#): the figures

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Change log

- 28 September 2026: first published.
 - 29 September 2026: version 2. Retitled from "Prague votes in rings, not along lines", which asserted a metro null the registered test could not establish; the kicker no longer says "pre-registered", the registration is described as "design written before modelling; not independently timestamped", and the earlier description of who reviewed the design is removed.
 - 29 September 2026: corrected "never half", which holds only for the registered block without age; with age added composition explains 52 %. H4a and H4b relabelled from "not supported" to "uninformative". "43 % is a lower bound" changed to "likely understated". Bootstrap groups corrected from 57 to 56 districts (rings_extended.json, G = 56); the effective sample is now stated beside the headline, with 49 districts for H3 and 44 for H4, from the same file.
 - 29 September 2026: cut the minor-party paragraph to one sentence (removed: the panel gradients of STAN and Stačilo!, the centre gradients of SPD and the Motorists, and SPD's metro interval, 0.02 to 0.17) and the uncited sentence on the metro as the city's spine. Merged the duplicated map description and question list; compressed the turnout section; moved the line D placebo to "What changed after registration"; added the contribution sentence, the further-checks table, the metadata block and the Reproduction section.
 - 29 September 2026: references renumbered by first citation, with access and snapshot dates; the Part 1 reference replaced by a link. Series renumbered: the parking study is Part 5.
 - 29 September 2026: added the section What this means.
 - Checked against editorial standard v1 on 29 September 2026.
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